

Logical Agent Case Studies – Complete Solutions

Case Study 1 — Solution

Define F_A =Fever, C_A =Cough, R_A =Rash, B_A =Breathlessness, Flu_A =Possible Flu, $Measles_A$ =Possible Measles, $Pneu_A$ =Possible Pneumonia. Rules: $F_A \wedge C_A \rightarrow Flu_A$; $F_A \wedge R_A \rightarrow Measles_A$; $C_A \wedge B_A \rightarrow Pneu_A$. Facts: F_A, C_A, B_A .

Forward chaining: $F_A + C_A$ fires Rule I $\rightarrow Flu_A$. Rule II cannot fire because R_A is absent. $C_A + B_A$ fires Rule III $\rightarrow Pneu_A$. Thus Flu and Pneumonia are derived.

Backward chaining: Goal $Pneu_A$ reduces through Rule III to C_A and B_A . Both are facts, so the goal succeeds.

Goal tree: $Pneu_A \rightarrow \{C_A \checkmark, B_A \checkmark\}$.

Case Study 2 — Solution

Rule 1 unifies with Ambulance using $\theta = \{x/Ambulance\}$; hence $ClearPath(Ambulance)$. Rule 2 gives $GreenSignal(Ambulance)$.

CNF: $C1 \neg Vehicle(x) \vee \neg EmergencyType(x) \vee ClearPath(x)$; $C2 \neg ClearPath(x) \vee GreenSignal(x)$; $C3 \neg ClearPath(x) \vee \neg RedSignal(x)$; $C4 \neg Vehicle(x) \vee \neg Behind(x,y) \vee \neg GreenSignal(y) \vee Proceed(x)$. Facts: $Vehicle(Ambulance)$, $EmergencyType(Ambulance)$, $Vehicle(CarA)$, $Behind(CarA,Ambulance)$. Negated goal: $\neg Proceed(CarA)$.

Resolution: $C1$ with the two Ambulance facts $\rightarrow ClearPath(Ambulance)$. $C2$ with that result $\rightarrow GreenSignal(Ambulance)$. $C4$ with $Vehicle(CarA)$, $Behind(CarA,Ambulance)$, $GreenSignal(Ambulance)$, using $\theta = \{x/CarA, y/Ambulance\}$, $\rightarrow Proceed(CarA)$. $Proceed(CarA)$ with $\neg Proceed(CarA) \rightarrow \blacksquare$.

Two emergencies can be independently inferred because rules are universal, but safe simultaneous coordination needs additional priority, intersection, route-conflict, and emergency-vehicle facts/rules.

Case Study 3 — Solution

CNF: $P1$ becomes $\neg SoilDry \vee IrrigationNeeded$. $P2$ becomes $(\neg IrrigationNeeded \vee ApplyDripMethod) \wedge (\neg CropWheat \vee ApplyDripMethod)$. $P3$ becomes $(ApplyDripMethod \vee CropAtRisk) \wedge (\neg CropWheat \vee CropAtRisk)$. Facts: $SoilDry$, $CropWheat$.

Goal $ApplyDripMethod$; add $\neg ApplyDripMethod$. Resolution: $(\neg SoilDry \vee IrrigationNeeded) + SoilDry \rightarrow IrrigationNeeded$. Then $(\neg IrrigationNeeded \vee ApplyDripMethod) + IrrigationNeeded \rightarrow ApplyDripMethod$. $ApplyDripMethod + \neg ApplyDripMethod \rightarrow \blacksquare$. Hence the theorem is proved.

If $SoilDry$ is removed, $IrrigationNeeded$ is not derivable, so $ApplyDripMethod$ is not entailed. $CropAtRisk$ also does not follow because $P3$ needs $\neg ApplyDripMethod$, and absence of $ApplyDripMethod$ is not the same as its negation.

Case Study 4 — Solution

Belief state starts with Stench[1,2], Breeze[1,1], Glitter[2,2]. Modus Ponens derives WumpusAdjacent[1,2], PitAdjacent[1,1], and Gold[2,2]. Rule 4 cannot derive Safe because no explicit $\neg$ Wumpus and $\neg$ Pit facts are supplied.

Forward chaining identifies possible Wumpus cells adjacent to [1,2]: [1,1], [1,3], [2,2] on a standard grid. Possible Pit cells adjacent to [1,1]: [1,2], [2,1]. These are possibilities, not exact locations.

The path [1,1]→[1,2]→[2,2] is NOT logically guaranteed safe. The KB cannot prove Safe[1,2] or Safe[2,2], and [1,2] is associated with a Stench percept. Therefore the agent should treat the path as uncertain/risky.